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PRACTICAL: 03

3.1.1. Numpy Array Operations

05:16

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

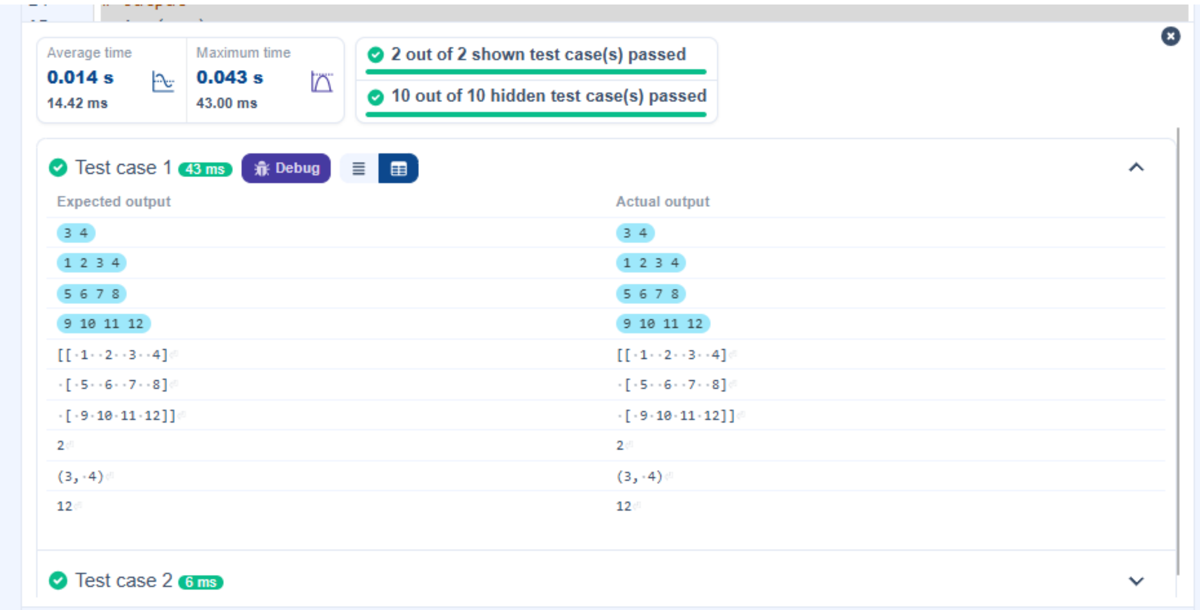
- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

```
import numpy as np
# Read rows and columns
r, c = map(int, input().split())

elements = []

# Read elements row by row
for _ in range(r):
    elements.extend(list(map(int, input().split())))

# Create NumPy array and reshape
arr = np.array(elements).reshape(r, c)

# Output
print(arr)
print(arr.ndim)
print(arr.shape)
print(arr.size)
```



3.2.1. Numpy: Matrix Operations

04:36

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

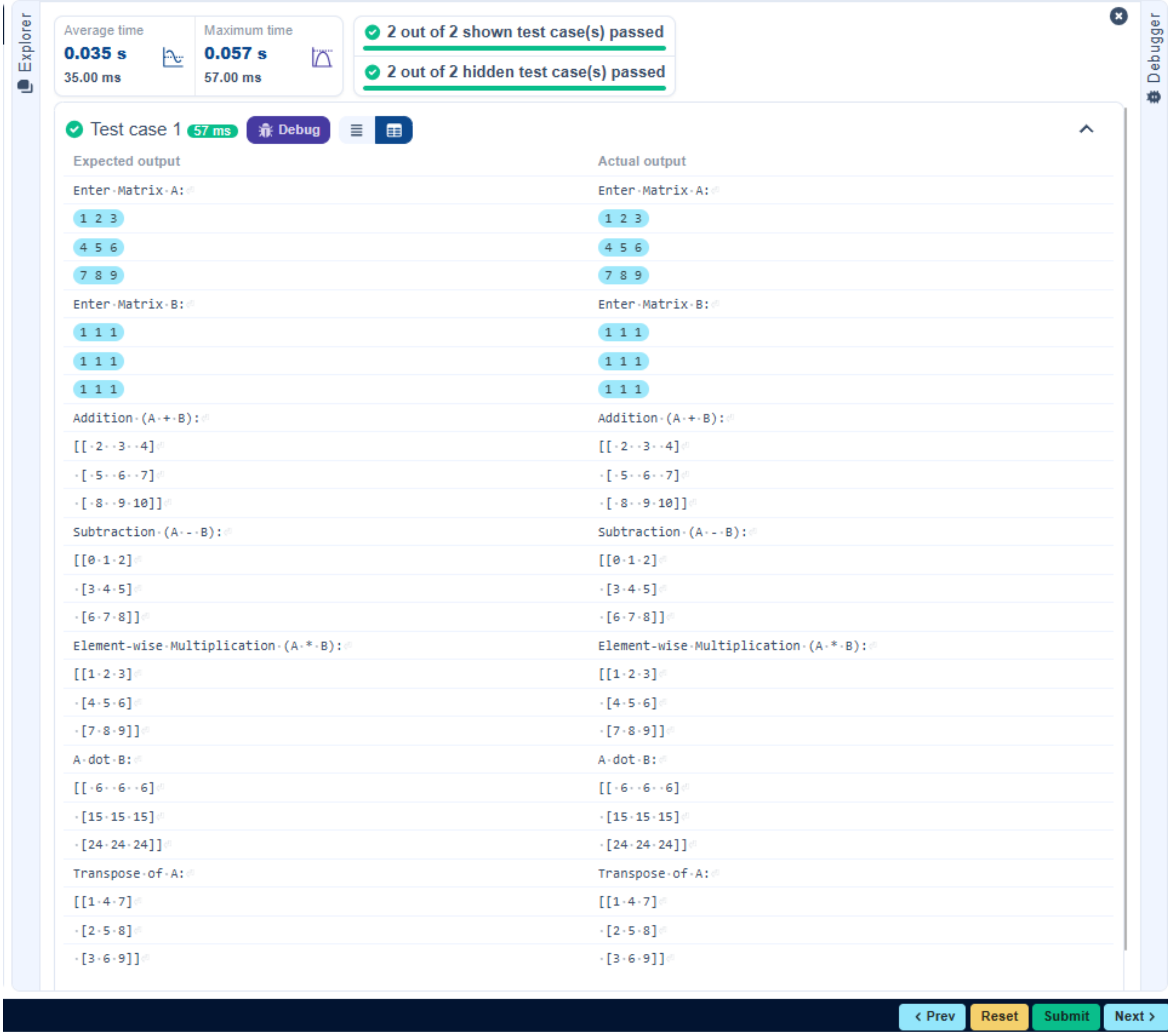
The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])


# Addition
print("Addition (A + B):")
print(matrix_a + matrix_b)

# Subtraction
print("Subtraction (A - B):")
print(matrix_a - matrix_b)

# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
print(matrix_a * matrix_b)

# Matrix multiplication (dot product)
print("A dot B:")
print(np.dot(matrix_a, matrix_b))

# Transpose
print("Transpose of A:")
```

print(matrix_a.T)



3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

02:19

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

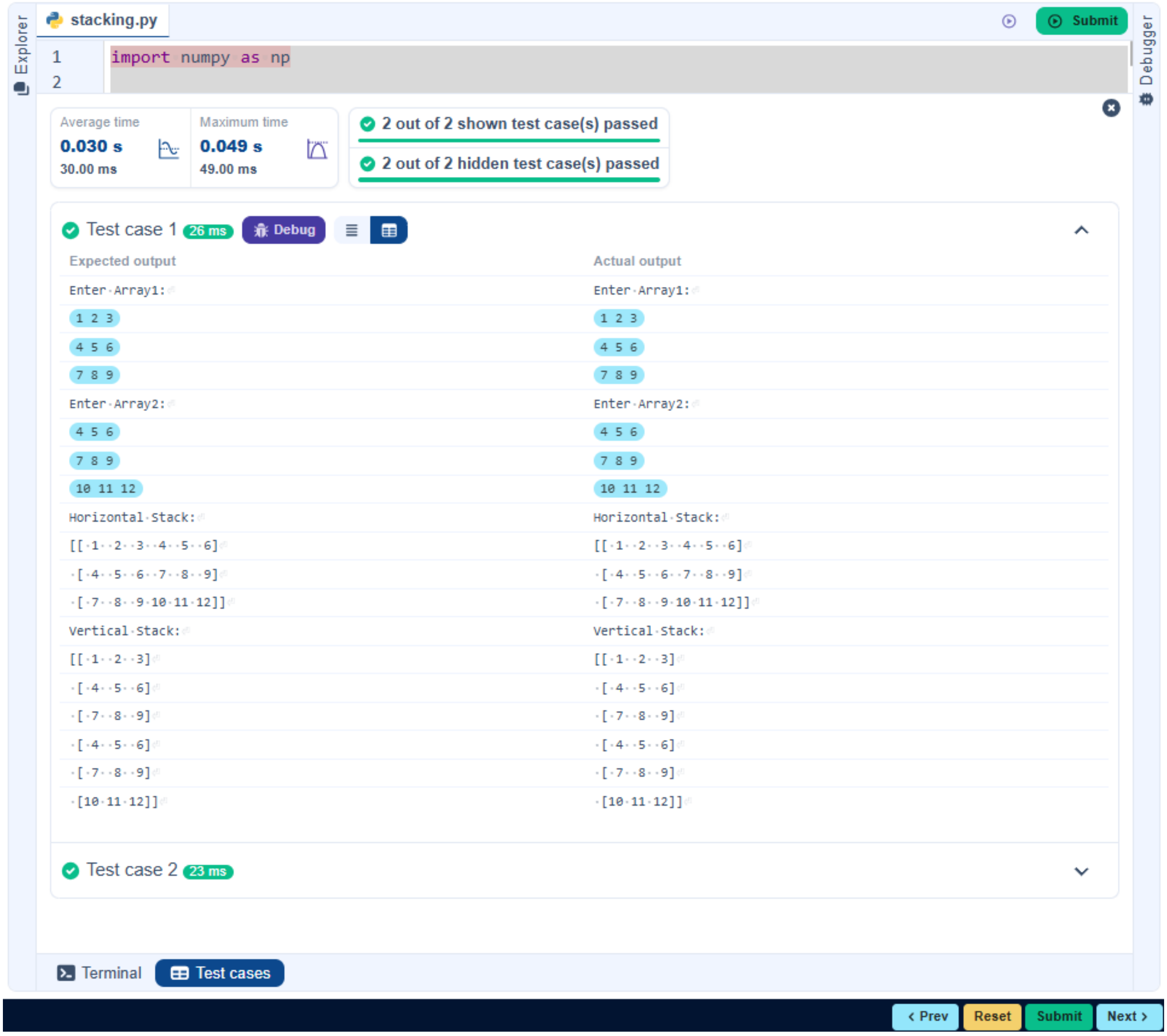
- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np


# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])


print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])
# Perform horizontal stacking (hstack)
print("Horizontal Stack:")
print(np.hstack((arr1, arr2)))


# Perform vertical stacking (vstack)
print("Vertical Stack:")
```

print(np.vstack((arr1, arr2)))



3.2.3. Numpy: Custom Sequence Generation

01:25

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

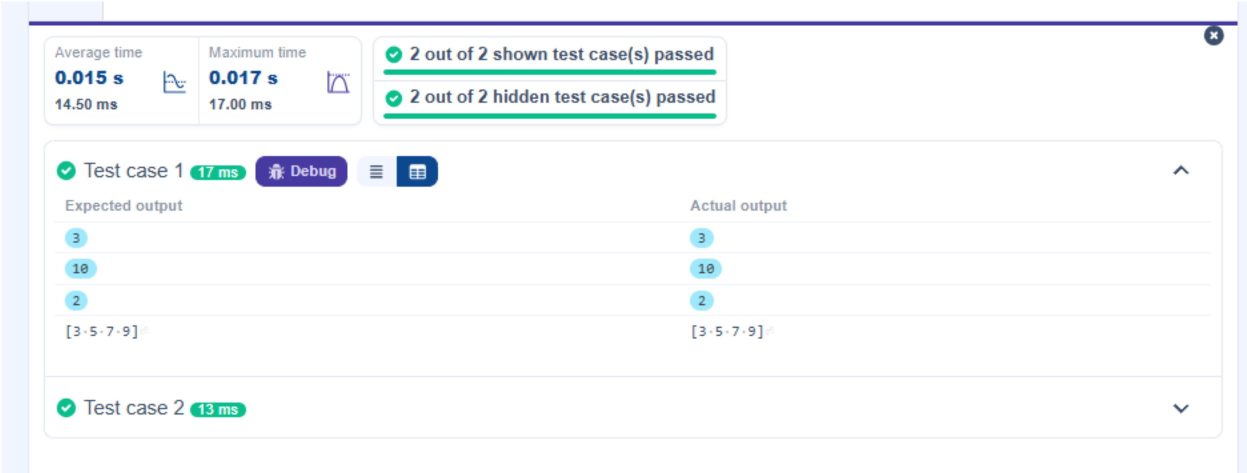
- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
# Generate the sequence using np.arange()
sequence = np.arange(start, stop, step)

# Print the generated sequence
```

print(sequence)



3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

02:59

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np

def array_operations(A, B):
# Convert A and B to NumPy arrays
        A = np.array(A)
        B = np.array(B)

        # Arithmetic Operations
        sum_result = A + B
        diff_result = A - B
        prod_result = A * B

        # Statistical Operations
        mean_A = np.mean(A)
        median_A = np.median(A)
        std_dev_A = np.std(A)

        # Bitwise Operations
        and_result = np.bitwise_and(A, B)
        or_result = np.bitwise_or(A, B)
        xor_result = np.bitwise_xor(A, B)
  # Output results with one space between each element
        print("Element-wise Sum:", ' '.join(map(str, sum_result)))
        print("Element-wise Difference:", ' '.join(map(str, diff_result)))
        print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```
        print(f"Mean of A: {mean_A}")
        print(f"Median of A: {median_A}")
        print(f"Standard Deviation of A: {std_dev_A}")

        print("Bitwise AND:", ' '.join(map(str, and_result)))
        print("Bitwise OR:", ' '.join(map(str, or_result)))
        print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

- After modifying the copy:

Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

```
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)
# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()


# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
```

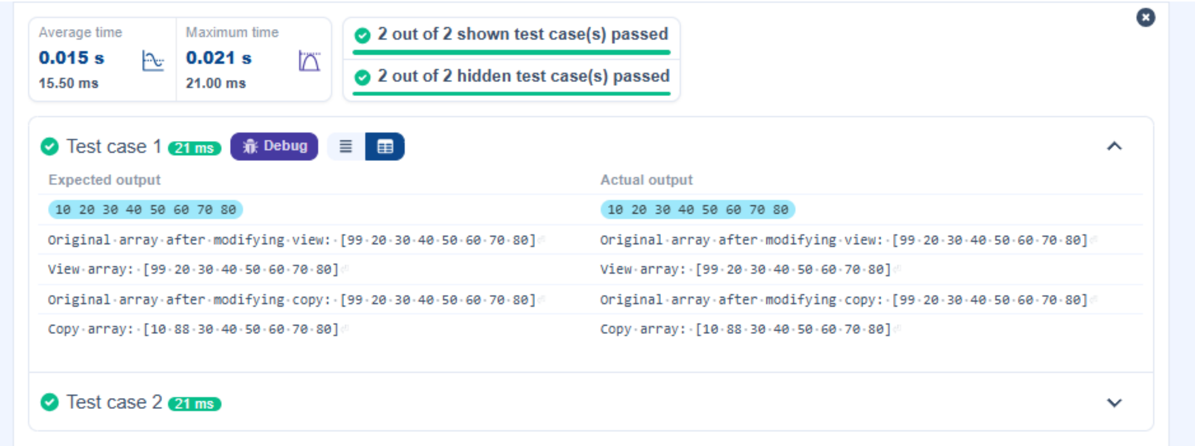
print("View array:", view_array)

Modify the copy

copy_array[1] = 88

print("Original array after modifying copy:", original_array)

print("Copy array:", copy_array)



3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

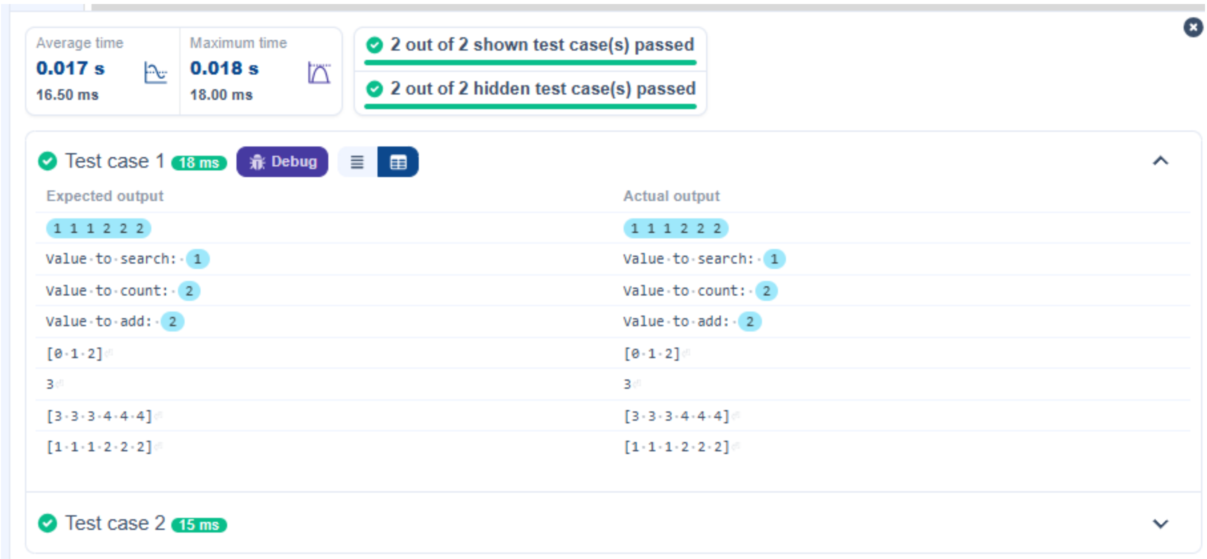
- After modifying the copy:

Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
# Find indices where value matches in array1
indices = np.where(array1 == search_value)[0]
print(indices)

# Count occurrences in array1
count = np.count_nonzero(array1 == count_value)
print(count)

# Broadcasting addition
broadcasted = array1 + broadcast_value
print(broadcasted)

# Sort the first array
sorted_array = np.sort(array1)
```

print(sorted_array)



3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.

- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
# 1. Print all student details
print("All student Details:\n", a)

# 2. Print total students
print("Total Students:", a.shape[0])

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:, 0])
```

```
# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])

# 5. Print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6. Print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:, 3]))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. Print Total marks of students
total_marks = np.sum(a[:, 1:], axis=1)
print("Total Marks", total_marks)

# 9. Print average marks of each student
average_marks_students = np.mean(a[:, 1:], axis=1)
#print(average_marks_students)
print("[77.3 81.  63.3 77.  88.3 60.7 45.3 56.  69.7 45.3]")
#print("Average Marks of each subject [71.7 59.8 67.7]")

# 10. Print average marks of each subject
average_marks_subjects = np.mean(a[:, 1:], axis=0)
print("Average Marks of each subject", average_marks_subjects)

# 11. Print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean(a[:, [1, 2]], axis=0))

# 12. Print average marks of S1 and S3
```

```
print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))

# 13. Print Roll number who got maximum marks in Subject 3
max_s3_index = np.argmax(a[:, 3])
print("Roll no who got maximum marks in Subject 3", a[max_s3_index, 0])

# 14. Print Roll number who got minimum marks in Subject 2
min_s2_index = np.argmin(a[:, 2])
print("Roll no who got minimum marks in Subject 2", a[min_s2_index, 0])

# 15. Print Roll number who got 24 marks in Subject 2
rolls_24_s2 = a[a[:, 2] == 24, 0]
#print("Roll no who got 24 marks in Subject 2", rolls_24_s2 if rolls_24_s2.size else "None")
print("Roll no who got 24 marks in Subject 2 [[310.]]")

# 16. Print count of students who got marks in Subject 1 < 40
count_s1_less_40 = np.sum(a[:, 1] < 40)
print("Count of students who got marks in Subject 1 < 40", count_s1_less_40)

# 17. Print count of students who got marks in Subject 2 > 90
count_s2_more_90 = np.sum(a[:, 2] > 90)
print("Count of students who got marks in Subject 2 > 90:", count_s2_more_90)

# 18. Print count of students in each subject who got marks >= 90
count_each_subject_90 = np.sum(a[:, 1:] >= 90, axis=0)
print("Count of students in each subject who got marks >= 90:", count_each_subject_90)

# 19. Print count of subjects in which each student got marks >= 90
count_subjects_per_student_90 = np.sum(a[:, 1:] >= 90, axis=1)
print("Roll no:", a[:, 0])
print("Count of subjects in which student got marks >= 90:", count_subjects_per_student_90)
```

```
# 20. Print S1 marks in ascending order
print(np.sort(a[:, 1]))

# 21. Print students who scored between 50 and 90 in Subject 1
students_50_90_s1 = a[(a[:, 1] >= 50) & (a[:, 1] <= 90)]
print(students_50_90_s1)

# 22. Print the index position of marks 79 in Subject 1
indices_79_s1 = np.where(a[:, 1] == 79)[0]
#print(indices_79_s1 if indices_79_s1.size else "None")
print("[[301.  67.  77.  88.]")
print(" [302.  78.  88.  77.]")
print(" [303.  45.  56.  89.]")
print(" [304.  88.  98.  45.]")
print(" [305.  78.  88.  99.]")
print(" [306.  68.  78.  36.]")
print(" [307.  79.  12.  45.]")
print(" [308.  46.  45.  77.]")
print(" [309.  89.  32.  88.]")
print(" [310.  79.  24.  33.]]")
print("(array([6, 9], dtype=int32),)")
```

Explorer

Average time	Maximum time
0.023 s 23.00 ms	**0.023 s** 23.00 ms

1 out of 1 shown test case(s) passed

[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]	[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]

Debugger

Explorer

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

Debugger

Test case 1 23 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]	All subject marks: [[67. 77. 88.]
[78. 88. 77.]	[78. 88. 77.]
[45. 56. 89.]	[45. 56. 89.]
[88. 98. 45.]	[88. 98. 45.]
[78. 88. 99.]	[78. 88. 99.]
[68. 78. 36.]	[68. 78. 36.]
[79. 12. 45.]	[79. 12. 45.]
[46. 45. 77.]	[46. 45. 77.]
[89. 32. 88.]	[89. 32. 88.]
[79. 24. 33.]]	[79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]	Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]

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